LocalCloudManager

System Design Description

Abstract
This document describes the design and implementation of the LocalCloudManager support system of Arrowhead 5.2.2 (`ah5-LocalCloudManager-java-spring`).

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1 Overview

This document describes the LocalCloudManager system implementation, which realizes the LocalCloudManager SysD [1]. LocalCloudManager is the operator console of the local cloud: it shows the state of the cloud and carries out the management actions an operator starts, with its own Arrowhead identity. In Section 2, we describe implementation details of the system.

2 Implementation

This implementation is based on the SysD document LocalCloudManager [1] and is located in `ah5-LocalCloudManager-java-spring` of the primary Java source tree (`Eclipse-Arrowhead-java`). It is the non-Java exception of the Java support distribution: a Node.js/nginx console without a JAR.

2.1 Implementation language and tools

- Node.js 20 (API, `amqplib`), nginx (UI), Node.js Docker Engine proxy.
- Storage: none.

2.2 Functional properties implementation

The following components implement the system:

Component	Responsibility
`api/src/server.js`	Routes, Arrowhead clients, registration.
`api/src/console-auth.js`	Console login and sessions.
`api/src/sysmonitor.js`	sysMonitor publisher.
`api/src/crl-status.js`	CRL loading and status for provider certificates.
`api/src/policies-api.js`	Local management-policy editor (ConsumerAuthorization grant/revoke with snapshot checks).
`api/src/rule-editor-api.js`	Rule-priority changes (native `modify-priorities` or create/remove replacement).
`api/src/sos-api.js`, `sos-model.js`	SoS compositions on PlantDescriptionEngine; validation, inclusion and cycle checks.
`api/src/cloud-config-api.js`	Pass-through to the LocalCloudConfigurator REST API.
`api/src/floorplans-api.js`	Chiplet floorplan view through orchestrated ChipletFloorplanGenerator services.
`ui/`	Browser console (nginx).
`docker-proxy/`	Docker Engine API proxy.

Every request except `GET /health` passes the proxy-secret check (only the console's nginx), the host/origin check, the console session check and, for changes, the confirmation check before the route handler calls the providing systems.

2.2.1 Data handled and stored

None durable; sessions and the identity token in memory. SoS compositions, policies and configurator designs are stored by their providing systems.

The interfaces exchange JSON objects built at run time; their structure is specified in the IDD documents.

2.3 Non functional properties implementation

2.3.1 Security

The console API has no Docker socket; Docker is reached only through the proxy on an internal network. The console is published on the host's loopback address only.

2.3.2 Power management

None; the system is a server process without power states. The `sysMonitor power` observation always reports line supply.

2.3.3 Internal monitoring

Logs go to stdout. The system publishes its `sysMonitor` observations every 30 s (see the sysMonitor IDD).

2.3.4 Configuration

Configuration is read from environment variables (listed in the SysD); secrets are read from files (`*_FILE` variables or mounted credential bundles).

2.4 Deployment

The local `compose.yaml` (project `ah5-Blocalcloudmanager`) builds and runs two containers from this folder: `cloud-monitor-api` (image `arrowhead-localcloudmanager-api-standalone:5.2.2`, port 8491, on the external cloud network and the internal `control` network) and `cloud-monitor-ui` (nginx, image `arrowhead-localcloudmanager-ui-standalone:5.2.2`, published on `127.0.0.1:8490`). Both run as UID 10001 with a read-only root filesystem, no capabilities and a non-executable `/tmp`. The identity password, console password hash and console proxy secret are Compose file secrets; the certificate, key, cloud CA and CRL directory are mounted read-only. The standalone definition starts no Docker proxy (`DOCKER_API_URL` points nowhere and `MANAGED_SERVICES` is empty), so the runtime view and container lifecycle are unavailable there; the configurator and floorplan views are disabled unless their variables are set. Copy `.env.example`, provision the identity, certificate and console secrets, then run `docker compose up -d -build` from this folder. ServiceRegistry, Authentication, DSO, ConsumerAuthorization and the grants of LocalCloudManager remain network dependencies.

The earlier combined test cloud (`Arrowhead-test-cloud-java`) is a separate deployment consumer; it ran the console with a Docker proxy as `cloud-Bmonitor-Bapi`, `cloud-Bmonitor-Bui` and `docker-Bproxy`.

2.5 Local modifications

Notes on this implementation in the workspace:

- Workspace system; operation names are camelCase (see the SD).
- Console endpoints for policies, rule priority, SoS, local-cloud configuration and chiplet floorplans are not part of the registered operation list.

2.6 Verification

`api/test`, `ui/test` and `docker-proxy/test` contain the console's unit tests (`node --test`): API routes, console authentication, CRL status, policies, rule priority, SoS, configurator and floorplan routes, and the UI models. The distribution report records 23 API tests for the packaged console.

3 Services

The implementation of the services is based on the SD and IDD documents referenced below.

Services produced	**SysD ref**	**SD ref**	**IDD ref**
`localCloudManagement`	[1]	[2]	[3]
`sysMonitor`	[1]	[4]	[5]
Services consumed	**SysD ref**	**SD ref**	**IDD ref**
serviceDiscovery; serviceRegistry Management	[6]	[7]	–
identity	[8]	[9]	–
serviceOrchestration; service OrchestrationHistoryManagement	[10]	[11]	–
serviceOrchestrationSimpleStore Management	[12]	[13]	–
authorizationManagement (query, grant, revoke policies)	[14]	[15]	–
dataHistorian	[16]	[17]	–
sysMonitorHistory; sysMonitor Management	[18]	[19]	–
plantDescriptionManagement	[20]	[21]	–
gatekeeperManagement; gate TunnelManagement	Gatekeeper, GateTunnel	–	–
Docker Engine subset	Docker proxy	–	–
configurator REST API	LocalCloud Configurator	–	–
floorplanDiscovery; floorplan Management	Chiplet Floorplan Generator	–	–

4 References

[1] J. Delsing, “LocalCloudManager – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-support-java-spring/LocalCloudManager/documentation/v5.2.2/LocalCloudManager-v5.2.2-SysD.pdf.

[2] ——, “localCloudManagement – Service Description (LocalCloudManager),” Eclipse Arrowhead, SD 5.2.2, 2026, localCloudManager-v5.2.2-localCloudManagement-SD.pdf.

[3] ——, “localCloudManagement HTTP/TLS/JSON – Interface Design Description (LocalCloudManager),” Eclipse Arrowhead, IDD 5.2.2, 2026, localCloudManager-v5.2.2-localCloudManagement-HTTP-IDD.pdf.

[4] ——, “sysMonitor – Service Description (LocalCloudManager),” Eclipse Arrowhead, SD 5.2.2, 2026, localCloudManager-v5.2.2-sysMonitor-SD.pdf.

[5] ——, “sysMonitor AMQP/TLS/JSON – Interface Design Description (LocalCloudManager),” Eclipse Arrowhead, IDD 5.2.2, 2026, localCloudManager-v5.2.2-sysMonitor-AMQP-IDD.pdf.

[6] ——, “ServiceRegistry – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-core-java-spring/serviceregistry/documentation/v5.2.2/ServiceRegistry-v5.2.2-SysD.pdf.

[7] ——, “serviceDiscovery – Service Description (ServiceRegistry),” Eclipse Arrowhead, SD 5.2.2, 2026, ah5-core-java-spring/serviceregistry/documentation/v5.2.2/ServiceRegistry-v5.2.2-serviceDiscovery-SD.pdf.

[8] ——, “Authentication – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-core-java-spring/authentication/documentation/v5.2.2/Authentication-v5.2.2-SysD.pdf.

[9] ——, “identity – Service Description (Authentication),” Eclipse Arrowhead, SD 5.2.2, 2026, ah5-core-java-spring/authentication/documentation/v5.2.2/Authentication-v5.2.2-identity-SD.pdf.

[10] ——, “DynamicServiceOrchestration – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-core-java-spring/serviceorchestration-dynamic/documentation/v5.2.2/DynamicServiceOrchestration-v5.2.2-SysD.pdf.

[11] ——, “serviceOrchestration – Service Description (DynamicServiceOrchestration),” Eclipse Arrowhead, SD 5.2.2, 2026, ah5-core-java-spring/serviceorchestration-dynamic/documentation/v5.2.2/DynamicServiceOrchestration-v5.2.2-serviceOrchestration-SD.pdf.

[12] ——, “SimpleStoreServiceOrchestration – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-core-java-spring/serviceorchestration-simple/documentation/v5.2.2/SimpleStoreServiceOrchestration-v5.2.2-SysD.pdf.

[13] ——, “serviceOrchestrationSimpleStoreManagement – Service Description (SimpleStoreServiceOrchestration),” Eclipse Arrowhead, SD 5.2.2, 2026, ah5-core-java-spring/serviceorchestration-simple/documentation/v5.2.2/SimpleStoreServiceOrchestration-v5.2.2-serviceOrchestrationSimpleStoreManagement-SD.pdf.

[14] ——, “ConsumerAuthorization – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-core-java-spring/consumer-authorization/documentation/v5.2.2/ConsumerAuthorization-v5.2.2-SysD.pdf.

[15] ——, “authorizationManagement – Service Description (ConsumerAuthorization),” Eclipse Arrowhead, SD 5.2.2, 2026, ah5-core-java-spring/consumer-authorization/documentation/v5.2.2/ConsumerAuthorization-v5.2.2-authorizationManagement-SD.pdf.

[16] ——, “DataManager – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-support-java-spring/DataManager/documentation/v5.2.2/DataManager-v5.2.2-SysD.pdf.

[17] ——, “dataHistorian – Service Description (DataManager),” Eclipse Arrowhead, SD 5.2.2, 2026, dataManager-v5.2.2-dataHistorian-SD.pdf.

[18] ——, “SysMonitor – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-support-java-spring/SysMonitor/documentation/v5.2.2/SysMonitor-v5.2.2-SysD.pdf.

[19] ——, “sysMonitorHistory – Service Description (SysMonitor),” Eclipse Arrowhead, SD 5.2.2, 2026, sysMonitor-v5.2.2-sysMonitorHistory-SD.pdf.

[20] ——, “PlantDescriptionEngine – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-support-java-spring/PlantDescriptionEngine/documentation/v5.2.2/PlantDescriptionEngine-v5.2.2-SysD.pdf.

[21] ——, “plantDescriptionManagement – Service Description (PlantDescriptionEngine),” Eclipse Arrowhead, SD 5.2.2, 2026, plantDescriptionEngine-v5.2.2-plantDescriptionManagement-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): location, new console modules, standalone deployment, consumed services and tests updated.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	